MIL-S-14623C(AL)
24 July 1985
SUPERSEDING
MIL-S-14623B(WC)
4 December 1974

MILITARY SPECIFICATION

SPRAY GUN, PAINT, LIGHT DUTY

This specification is approved for use by the US Army Armament Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of The Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one pressure feed type of air-operated, light duty, paint spray gun.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-12651 - Sprayers, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-ETS (R), Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-889 - Dissimilar Metals.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1), the contractor shall furnish one or more paint spray guns for first article inspection (see 4.3 and 6.2.1). The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design and construction. The spray gun shall be new and of the manufacturer's latest design which shall meet the requirements specified herein. The gun shall be designed for hand operation for limited and intermittent use. The spray gun shall be a pressure feed type bleeder. The gun shall have an internal or external mix air cap and shall include a fluid pressure cup. The design shall be such that parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement without modification, distortion, or damage to the gun or any component. The paint spray gun shall be furnished complete so that it can be used for the operations specified herein.

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* 3.2.1 Interchangeability. All parts shall be manufactured to definite standards, clearances, and tolerances in order that any such parts of a particular type or model may be replaced, interchanged, and adjusted without modification of the replacement parts or any other parts of the unit. When practical, all parts shall be permanently and legibly marked in accordance with MIL-STD-130.

* 3.2.2 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

* 3.2.3 Fastening devices. Screws, pins, bolts, and similar parts shall be installed with means for adjustment and for preventing loss of tightness. The methods for adjustment and for preventing loss of tightness shall be in accordance with accepted engineering standards and practices. All such parts when subject to renewal or adjustment shall not be swaged, peened, staked, or otherwise deformed.

* 3.2.4 Castings and forgings. Castings shall be free from blowholes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings and forgings shall be free from any properties adversely affecting their strength, durability, or suitability. Reclaiming or repair of major castings and forgings shall be permitted if the repair is not on a critical surface.

* 3.2.5 Welding, brazing, and soldering. Welding, brazing, and soldering shall be of a quality which shall sustain the performance requirements of the welded, brazed, or soldered part. These operations shall not be employed as repair measures for defective parts.

* 3.2.6 Surface. Surfaces of all castings, forgings, stampings, and parts shall be clean and free of sand, dirt, fins, sprues, scale, flux, and other harmful or extraneous materials.

* 3.2.7 Protective finishes. Exposed metal (see 6.4.1) and working surfaces, except when the parts involved are inherently corrosion resistant, shall be protected with an electrochemical or chemical conversion finish. The fluid nozzle covered in 3.5.2.3 is excluded from this requirement.

* 3.3 Material. Materials not specifically designated herein or in the contract shall be of a quality commensurate with commercial practice within the producing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or the overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

* 3.3.1 Threaded aluminum interfacing. Threaded aluminum or aluminum alloy parts or components of the spray gun or cups shall not be screwed, fastened, or affixed to other male or female threaded aluminum parts or components. Oxidation and corrosion of mating (male and female) threads and fasteners can provide for

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physical bonding of the aluminum alloy materials, which could interfere with, and prevent, disassembly and assembly of the spray gun.

* 3.3.2 Gaskets, O-rings, bushings, seals. The spray gun shall include gaskets, O-rings, bushings, or other seals to assure that there are no internal or external air, paint, or mixture leakages. All gaskets, O-rings, seals, and bushings contained in, or provided with, the spray gun shall be composed of materials which shall not undergo mechanical or chemical degradation after repeated and prolonged exposure to paints, coatings, thinners, and solvents. Non-mechanical seals such as glues, epoxies, and anaerobics shall not be used as sealants in the spray gun.

3.4 Performance and product characteristics.

3.4.1 Sprayable material. The gun shall spray enamels, lacquers, primers, shellacs, varnishes, stains, oil paints, and related fluids with a viscosity range of 16 to 26 seconds using a No. 4 Ford cup at +77° ±5°F. The gun shall satisfactorily spray these materials on vertical surfaces without spitting, splattering, fogging, or leakage.

3.4.2 Spray pattern. The spray gun shall be capable of producing a normal fan spray pattern. (See Figure 1, normal spray pattern.) The spray gun shall not produce any of the defective patterns shown in Figure 1. The spray pattern shall rotate but not change in shape when the air cap is rotated. The spray gun, using spray fluids complying with 3.4.1, shall produce a fan spray of minimum effective dimensions of two inches along the minor axis and 6 1/2 inches along the major axis when held stationary at a distance of eight inches from, and perpendicular to, the surface to be sprayed.

* 3.4.3 Air consumption. The spray gun shall operate satisfactorily without spitting, splattering, fogging, or leaking and shall produce the normal spray patterns specified in 3.4.2 with air consumption of two to eight standard cubic feet per minute (scfm) at a pressure of 20 to 40 pound-force per square inch gage (psig).

3.4.4 Operational endurance. The spray gun and all components shall withstand without damage and shall perform as specified in 3.4.1, 3.4.2, and 3.4.3 after 5,000 actuations (full stroke) of the spray gun trigger.

3.4.5 Tightness requirements.

3.4.5.1 High pressure effect. The spray gun and all components shall be capable of withstanding without damage an air pressure of 60 psig for a period of 30 seconds.

3.4.5.2 Fluid leakage. With the fluid needle valve in the normal position (closed), no indication of fluid leakage shall occur when subjecting the spray gun to a pressure of 40 psig for a five minute period.

3.5 Components. The spray gun shall consist of, but not be limited to, the principal components covered in 3.5.1 through 3.5.5.

3.5.1 Body. The spray gun body shall incorporate a head, a pistol grip handle, and a hanging hook. The head shall be accurately machined to provide

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support and correct alignment of the nozzle component and shall be an integral part of, or detachable from, the body. The pistol grip handle shall be designed for two-finger trigger control and shall fit the operator's hand with good balance. The hook shall be provided at the top of the body for hanging the spray gun when not in use. The body shall be made of aluminum alloy.

3.5.2 Nozzle assembly. The nozzle assembly shall consist, essentially, of an air cap, fluid needle valve, fluid nozzle, and the required auxiliary parts necessary for spraying the materials specified in 3.4.1. The nozzle shall be designed to embody the required performance and detailed characteristics of the spray gun specified herein. The nozzle shall be adjustable from a horizontal to a vertical spray pattern.

3.5.2.1 Air cap. The air cap shall provide the required air openings for complete atomization without spattering, fogging, or splitting the spray pattern specified in 3.4.2 using the air consumption specified in 3.4.3. The air cap shall be an external or internal mix type and shall be secured to the spray gun head by a retainer ring or other suitable means. The air cap shall be self-centering and shall be removable and replaceable to permit cleaning and replacement. The external mix-type air cap, if used, shall be fabricated of drawn steel, brass, or other suitable materials. The internal mix type air cap, if used, shall be of surface hardened steel. The materials used in both type air caps shall have the strength and physical properties to function in accordance with the performance requirements specified herein.

3.5.2.2 Fluid valve. The fluid valve shall be of the needle type and shall be operated (opened and closed) by operating the spray gun trigger. The fluid valve shall be held normally closed by a coil spring enclosed within the spray gun body to shut off the fluid flow. The fluid valve shall be in contact with an adjusting screw at the rear of the gun body above the handle grip which shall be used for changing (regulating) the quantity of fluid flow through the fluid orifice when the spray gun trigger is operated. The fluid needle valve shall be made of stainless steel. The valve packing shall be graphited fiber, treated leather, or synthetic material.

* 3.5.2.3 Fluid nozzle. The fluid nozzle shall be made of carbide or of steel, heat treated and hardened to an equivalent of 48-60 Rockwell C. All interior surfaces affecting material flow, such as the fluid orifice, and all surfaces affecting air flow shall be smooth and finished to meet the performance requirements specified herein.

3.5.2.4 Auxiliary parts. All baffle plates, tension washers, retainer rings, coil springs, and other attaching and auxiliary parts shall be of material capable of meeting the operational requirements specified herein.

3.5.3 Trigger. The trigger shall be designed for two-finger control. The trigger shall operate freely with a positive action and shall be of such design and location in front of the handle grip to hold the fingers in a normal position when using the spray gun in the spraying position. The trigger shall operate the fluid valve, but with air flow commencing before fluid flow. The trigger shall be fabricated of metal capable of withstanding the operational requirements specified herein.

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* 3.5.4 Air hose connection. The spray gun shall incorporate an air hose connection. Unless otherwise specified (see 6.2.1), the air hose connection shall be a 1/4-18 male standard straight National Pipe Thread (NPT), with a 60 degree bevel taper seat. The air hose connection shall be located at a position on the spray gun body that shall not interfere with the operation of the spray gun; however, the connection shall be accessible for connecting the air base.

3.5.5 Fluid cup. The spray gun shall be fitted with a pressure fluid cup with a minimum capacity of 28 ounces, mounted below the spray gun head. The cup cover shall be an integral part of, or detachable from, the spray gun body and shall provide a liquid and air tight seal with the spray gun head. The cup shall be of seamless aluminum construction with a minimum thickness of 0.040 inch. The outside diameter of the cup shall not be less than four inches and the throat diameter shall not be less than two inches. The cup shall be detachable from the cup cover. When assembled, the cup cover and cup shall be secure and provide a liquid and air tight seal.

3.6 Product identification. The spray guns shall be marked for product identification in accordance with MIL-STD-130 and, unless otherwise specified (see 6.2.1), shall include the National Stock Number.

* 3.7 Workmanship. Standards of workmanship shall assure that the spray guns shall have the stability, durability, safety, and efficient operating characteristics found in the best commercial units and as specified in Section 3.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Unless otherwise specified (see 6.2.1), quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (d) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

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(a) Product examination (see 4.5).
(b) Fluid leakage test (see 4.6.2).
(c) Spray pattern (performance) test (see 4.6.4).
(d) Packaging inspection (see 4.7).

* 4.4.1 Sampling. When required (see 6.2.1), sampling for quality conformance inspection shall be as specified in MIL-STD-105 at the Inspection Level and Acceptable Quality Level (AQL) specified in the contract. For packaging inspection (see 4.7), special inspection level S-4 and an AQL of 4.0 percent defective shall be used.

* 4.5 Product Examination. Visually and manually examine the spray gun to determine conformance with the requirements of 3.2 through 3.3.2 and 3.5 through 3.7. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the spray gun has passed the examinations prescribed for it by the contractor's inspection system shall be cause for rejection.

4.6 Tests.

4.6.1 Operational endurance test. The spray gun shall be subjected to 5,000 cycles of operation of the trigger. On conclusion of the test, the spray gun shall be examined for damage and tested for leakage and performance in accordance with 4.6.2 and 4.6.4, respectively, to determine compliance with 3.4.4.

4.6.2 Fluid leakage test. The spray gun shall be tested for fluid leakage by connecting an air hose line to the air inlet connection of the spray gun. The spray gun fluid cup shall be filled with a dark color fluid having a viscosity of 22 seconds using a No. 4 Ford cup at +77° ±5°F. An air pressure of 40 psig shall be applied to the spray gun. The pressure gage used shall have an accuracy within 0.25 percent. The spray gun trigger shall be operated several times to permit the fluid flow to reach the fluid tip while under an air pressure of 40 psig. With the fluid valve in the normal position (closed), the spray gun shall be permitted to blow against a white card, of a minimum size of ten inches by five inches, with the spray gun perpendicular to, and at a distance of eight inches from, the card. The spray gun shall be allowed to operate in this condition for a period of five minutes. During and upon conclusion of the test, the white card shall be observed for discoloration due to fluid leakage. The valve packing nut, the connection of the cup cover to the cup container, and any other connections in the path of the fluid flow shall be examined to determine compliance with 3.4.5.2.

4.6.3 High pressure test. The air ports in the fluid nozzle and air cap shall be blocked by fitting the spray gun with a blank fluid nozzle and a blank air cap, or by other suitable means. An air hose shall be connected between the spray gun air inlet connection and the air source and 60 psig pressure shall be exerted on the spray gun for a period of 30 seconds. The spray gun shall be examined to determine compliance with 3.4.5.1. The pressure gage used shall have an accuracy within 0.25 percent.

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4.6.4 Spray pattern (performance) test. The spray gun shall be performance tested with a pigment material having a viscosity of 16 seconds using a No. 4 Ford cup at +77° ±5°F. The spray gun shall be connected to an air source capable of supplying the required consumption specified in 3.4.3. The spray gun shall be held perpendicular to, and at a distance of not more than eight inches from, the vertical surface to be sprayed. The spray pattern shall conform to the requirements of 3.4.2. The spray gun shall be checked for producing horizontal and vertical patterns when the air cap is rotated. This test shall be repeated using a pigment material having a viscosity of 26 seconds using a No. 4 Ford cup at +77° ±5°F to determine compliance with 3.4.1 and 3.4.2.

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the requirements of Section 5.

5. PACKAGING

* 5.1 Military preservation, packing, and marking. When specified (see 6.2.1), requirements for Level A preservation and for Levels A and B packing and marking shall be in accordance with MIL-S-12651.

* 5.2 Industrial preservation, packing, and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D3951-82.

6. NOTES

6.1 Intended use. This specification covers a hand operated, light duty spray gun intended for use in the application of spray material by compressed air. The spray gun is intended for limited and intermittent duty for producing a fine finish on various surfaces.

6.2 Ordering data.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Air hose connection if different (see 3.5.4).
(d) Product identification if different (see 3.6).
(e) First article inspection if different (see 4.3).
(f) Quality conformance inspection if different (see 4.4).
(g) Sampling when required (see 4.4.1).
(h) Level of preservation, packing, and marking if different (see 5.1 and 5.2).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on a DD Form 1423 incorporated in the contract.

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* 6.4 Definitions.

6.4.1 Exposed metal. Exposed metal is defined as metal exposed to atmospheric conditions, compressed air, solvents, and the material (paint) being sprayed (see 3.2.7).

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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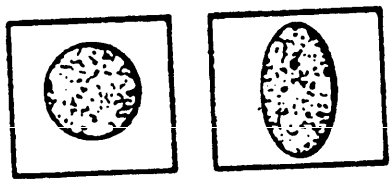
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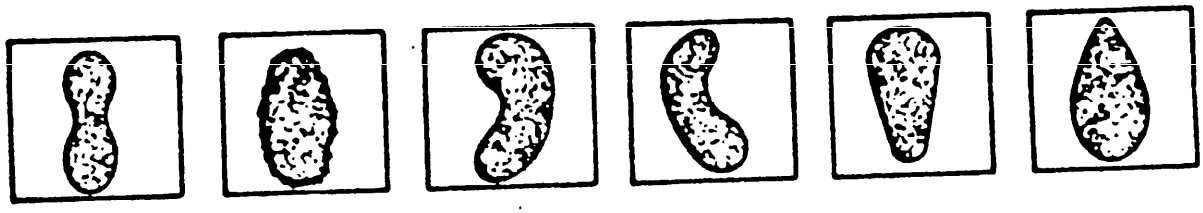
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NORMAL SPRAY PATTERNS



DEFECTIVE SPRAY PATTERNS

FIGURE 1. Spray patterns.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

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MIL-S-14623(AL), Rev C	Spray Gun, Paint, Light Duty

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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